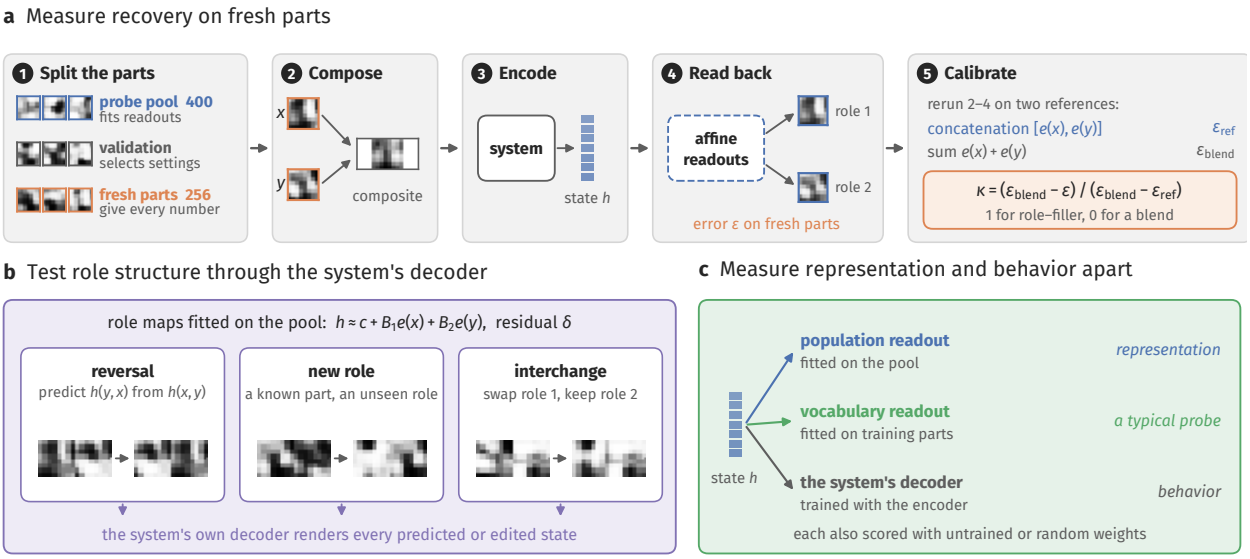

a Measure recovery on fresh parts
1 Split the parts
probe pool 400
fits readouts
validation
selects settings
fresh parts 256
give every number
2 Compose
x
y
composite
3 Encode
system
state h
4 Read back
affine readouts
role 1
role 2
error ε on fresh parts
5 Calibrate
rerun 2–4 on two references:
concatenation [e(x), e(y)]
ε_ref
sum e(x) + e(y)
ε_blend
κ = (ε_blend − ε) / (ε_blend − ε_ref)
1 for role–filler, 0 for a blend
b Test role structure through the system's decoder
role maps fitted on the pool: h ≈ c + B1e(x) + B2e(y), residual δ
reversal
predict h(y, x) from h(x, y)
new role
a known part, an unseen role
interchange
swap role 1, keep role 2
the system's own decoder renders every predicted or edited state
c Measure representation and behavior apart
population readout
fitted on the pool
representation
vocabulary readout
fitted on training parts
a typical probe
state h
the system's decoder
trained with the encoder
behavior
each also scored with untrained or random weights